

Flexural Analysis of Laminated CFRP Beams Under Four-Point Bending

**Bachelor Thesis Project II (BTP-II) by
Nagoju Hemanth Sri Sai Harshith
(22AE30036)
Under the supervision of
Prof. Anup Ghosh**



Introduction

- Laminated carbon fiber reinforced polymer (CFRP) beams are widely used in aerospace and structural applications due to their high strength-to-weight ratio.
- Unlike any other materials, CFRP exhibits anisotropic behavior, meaning its properties depend on fiber orientation.
- The four-point bending test is commonly used because it provides a pure bending region, allowing accurate evaluation of flexural properties.

Problem Statement

- The flexural behavior of laminated composites is complex due to variation in fiber orientation and layer interaction.
- Traditional beam theory is not sufficient to fully describe the behavior of such materials.
- A combined approach using theoretical models and numerical simulations is required to accurately predict stress, strain, and deflection.

Objectives

- To study the flexural behavior of Carbon Fiber beams under different loading conditions using a four-point bending setup
- To analyze the influence of different laminate configurations on stiffness and deformation.
- To calculate key parameters such as maximum stress, strain, and deflection.
- To validate the numerical results using Classical Laminate Theory (CLT)

Material Properties

- composites offer excellent mechanical properties such as high tensile strength (600–1500 MPa) and high stiffness (70–150 GPa).
- They are lightweight, corrosion-resistant, and exhibit superior fatigue performance.
- Due to their layered structure, their mechanical behavior varies depending on fiber orientation and stacking sequence.

Classical Laminate Theory (CLT)

Classical Laminate Theory (CLT) is used to analyze the bending behavior of composite laminates.

It treats the laminate as a stack of orthotropic plies, each contributing to overall stiffness.

The relationship between forces, moments, strains, and curvatures is expressed using stiffness matrices.

Governing Equation:

$$\begin{bmatrix} N \\ M \end{bmatrix} = \begin{bmatrix} A & B \\ B & D \end{bmatrix} \begin{bmatrix} \varepsilon_0 \\ \kappa \end{bmatrix}$$

A = Extensional stiffness matrix, **B** = Coupling matrix (zero for symmetric laminates), **D** = Bending stiffness matrix

CLT – Bending & Strain Relations

Bending Relation (Pure Bending):

$$M = D\kappa$$

This replaces classical $M=EI/R$ for composites.

Bending stiffness depends on laminate stacking sequence

Stress in Each Ply:

$$\sigma^{(k)} = \bar{Q}^{(k)} \varepsilon^{(k)}$$

$\bar{Q}^{(k)}$ → Transformed stiffness matrix of kth ply

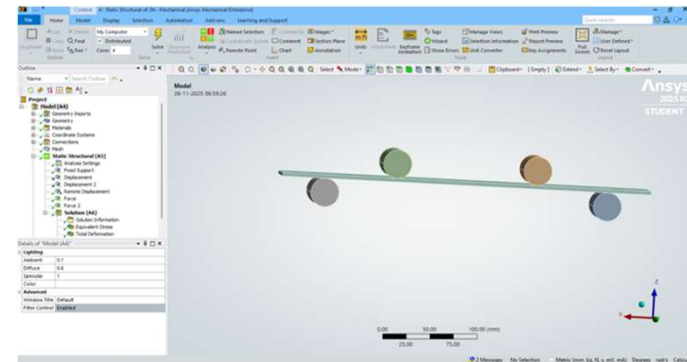
Captures effect of fiber orientation

Beam Configuration

- The beam is modeled as a simply supported laminated structure subjected to four-point bending.
- Two equal loads are applied symmetrically at one-third of the span.
- This setup creates a constant bending moment region in the center of the beam, decreasing shear effects.
- Laminate configurations:
 - Four different stacking sequences are considered to study the effect of fiber orientation:
 - $[0^8]_t$, $[90^8]_t$, $[(\pm 45)^4]_t$, $[(0/90)^4]_t$
 - These configurations help compare stiffness and deformation behavior.

ANSYS Modelling

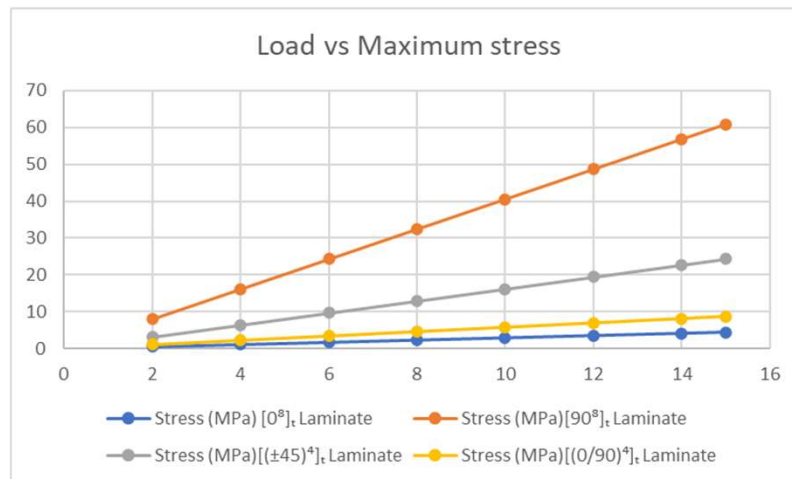
- A three-dimensional finite element model is developed using ANSYS 2026 R2.
- Each ply is assigned orthotropic material properties based on Carbon fiber characteristics.
- The model is discretized into small elements to accurately capture stress and strain variations.
- Simply supported boundary conditions are
- applied at both ends of the beam.
- Load cases range from 2 N to 15 N to study behavior
- under increasing loading conditions.



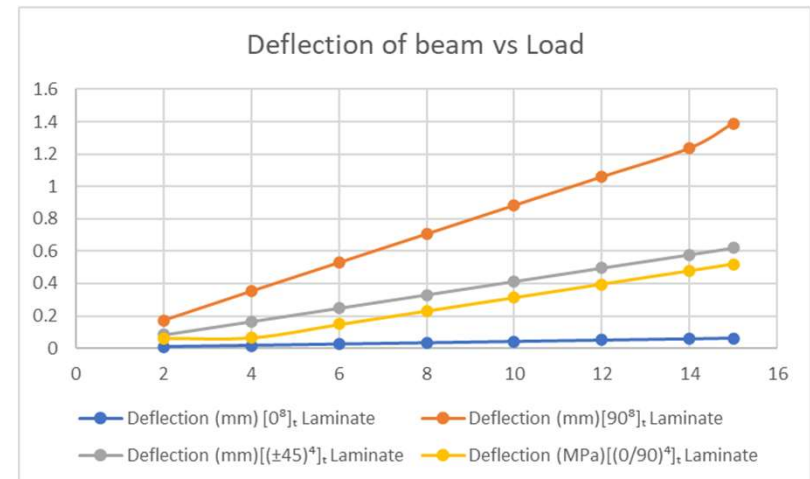
Parameters Extracted

- The analysis focuses on extracting key structural responses such as:
- Mid-span deflection
- Maximum stress in laminate
- Strain distribution across layers
- These parameters are used to compare numerical results with theoretical predictions.

Results

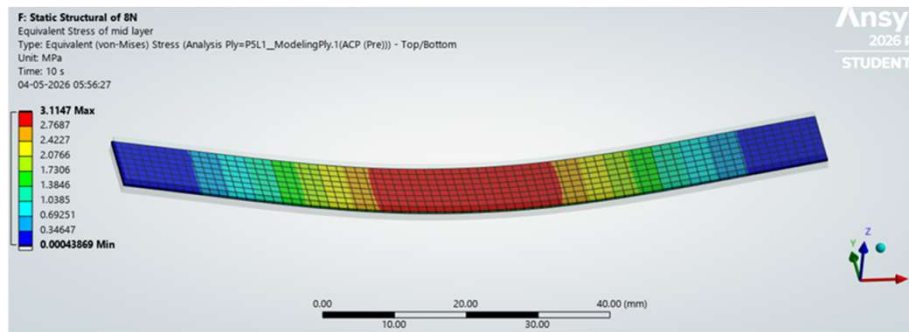


Plot of maximum stress in laminate for different loads

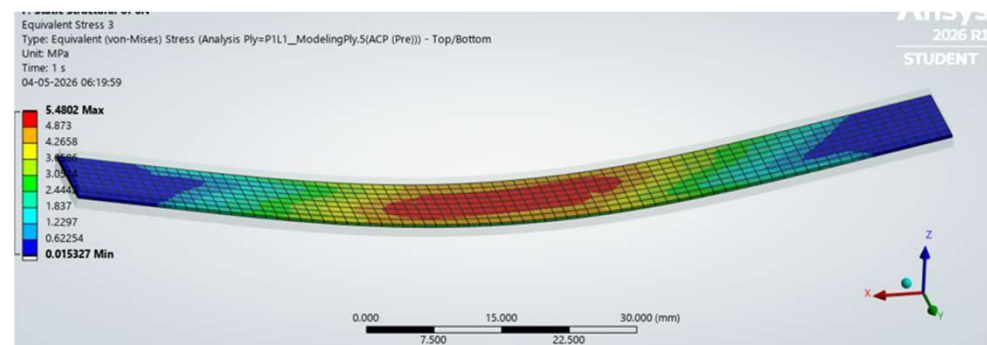


Plot of deflection in beam for different loads

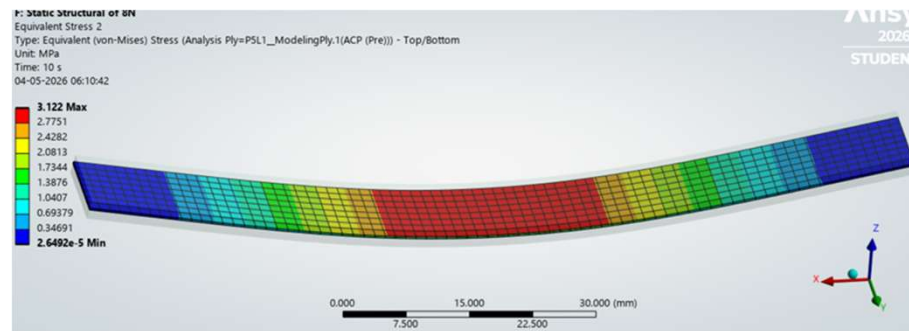
Results



Stress contour for the middle layer of $[0^8]_t$ laminate

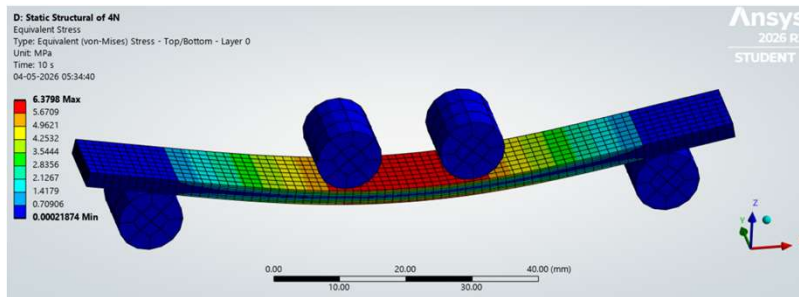


Stress contour for the middle layer of $[(45/-45)^4]_t$ laminate

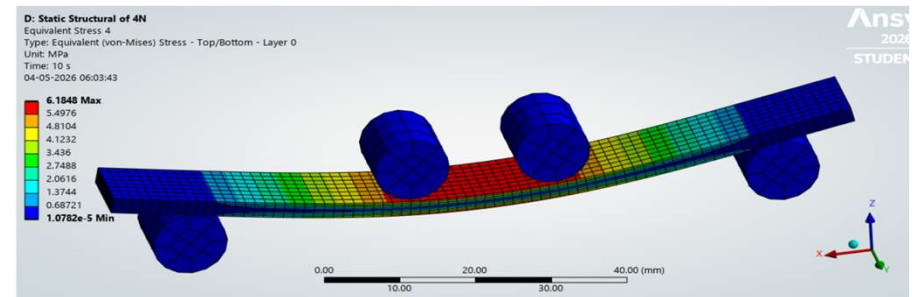


Stress contour for the middle layer of $[90^8]_t$ laminate

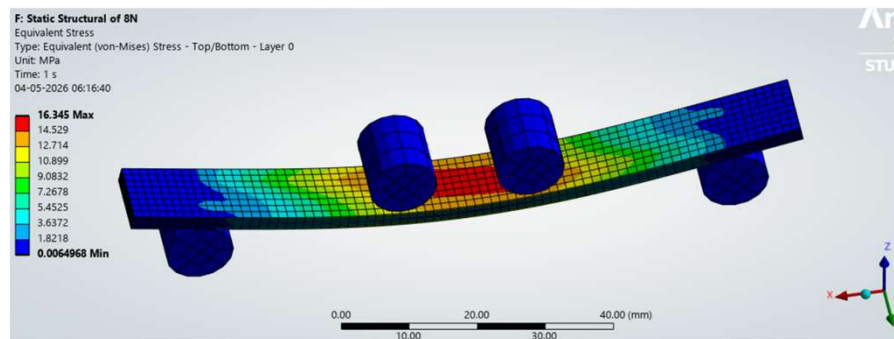
Results



Stress contour at 4 N Load for $[0^8]_t$ laminate



Stress contour at 4N Load for $[90^8]_t$ laminate



Stress contour at 8 N Load for $[(45/-45)^4]_t$ laminate

Comparison of Laminates

- The $[0^8]_t$ laminate exhibits the highest stiffness due to fibers aligned with the loading direction.
- The $[90^8]_t$ laminate shows the lowest stiffness and highest deflection.
- The $[(0/90)^4]_t$ laminate provides a balanced response.
- The $[(\pm 45)^4]_t$ laminate shows moderate stiffness with noticeable shear effects.

Conclusion

- The flexural behavior of CFRP beams is highly dependent on stacking sequence and fiber orientation.
- All laminate configurations showed linear elastic behavior within the applied load range.
- The $[0^8]_t$ laminate is the most efficient in resisting bending, while $[90^8]_t$ is the least stiff.
- Finite element results were successfully validated using Classical Laminate Theory.
- The use of epoxy carbon prepreg with very high modulus (395 GPa) significantly enhances bending stiffness.

Future Work

- Extend the analysis to higher load levels to investigate nonlinear behavior, yielding, and failure mechanisms such as fiber breakage and delamination.
 - Study the influence of different stacking sequences and ply thickness variations to optimize stiffness and strength.
 - Include environmental effects such as temperature, moisture absorption, and fatigue loading for real-world applications.
 - Extend the study to hybrid composites or sandwich structures to explore improved structural performance.
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THANK YOU

